

```
# cspassword is a valid value for this variable.
$db_driver = 'mysql';

# the db name or IP address of the host that the database server runs on.
$db_host = 'localhost';

# the name of the database. For MySQL, this is the database's SID. For a
# SQLiTP, this is a name (as path) for the db file.
$db_name = 'bugzilla';

# also we connect to the database as:
$db_user = 'root';
```

Figure 1: Configuring Bugzilla by changing the *localconfig* file

```
'password';
> GRANT ALL PRIVILEGES ON *.* TO 'bugzilla'@'localhost';
> FLUSH PRIVILEGES;
mysql > CREATE DATABASE bugzilla_db CHARACTER SET utf8;
> GRANT SELECT,INSERT,UPDATE,DELETE,INDEX,ALTER,CREATE,DROP,
REFERENCES ON bugzilla_db.* TO 'bugzilla'@'localhost'
IDENTIFIED BY 'cspassword';
> FLUSH PRIVILEGES;
> QUIT
```

- Use the following command to connect the user with the database:

```
$mysql -u bugzilla -p bugzilla_db
$mysql > use bugzilla_db
```

Step 3: Bugzilla installation and configuration

After downloading the Bugzilla archive from the URL mentioned above, untar the package into the */var/www* directory. All the configuration related information can be modified by the *localconfig* file. To start with, set the variable *\$webservergroup* as 'www' and set other items as mentioned in Figure 1.

Followed by the configuration, installation can be completed by executing the following Perl script. Ensure this script is executed with root privileges:

```
$ ./checksetup.pl
```

Step 4: Integrating Bugzilla with Apache

Insert the following lines in the Apache server configuration file (*server.xml*) to integrate Bugzilla into it. Place the directory bugzilla inside *www* in our build folder:

```
<Directory /var/www/bugzilla>
AddHandler cgi-script.cgi
Options +ExecCGI
DirectoryIndex index.cgi index.html
AllowOverride Limit FileInfo Indexes Options
</Directory>
```

Our set up is now ready. Let's hit the address in the browser to see the home page of our freshly deployed Web application (<http://localhost/bugzilla>).



Figure 2: Bugzilla main page



Figure 3: Defect lifecycle



Figure 4: New account creation

Defect lifecycle management

The main purpose of Bugzilla is to manage the defect's lifecycle. Defects are created and logged in various phases of the project (e.g., functional testing), where they are created by the test engineer and assigned to development engineers for resolution. Along with that, managers or team members need to be aware of the change in the state of the defect to ensure that there is a good amount of traceability of the defects. When the defect is created, it is given a 'new' state, after which it is assigned to a development engineer for resolution. Subsequently, it will get 'resolved' and eventually be moved to the 'closed' state.

Step 1: User account creation

To start using Bugzilla, various user accounts have to be created. In this example, Bugzilla is deployed in a server named 'hydrogen'. On the home page, click the 'New Account' link available in the header/footer of the pages (refer to Figure 4). You will be asked for your email address; enter it and click the 'Send' button. After registration is accepted, you should receive an email at the address you provided confirming your registration. Now all you need to do is to